

```

fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
% 2 primary fits 1 for 110 and another one for 200 tangential
% directions:
fit_data_file = 'FitEn_cut400ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.
fit_data_file = 'FitEn_cut400ref200_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.

if ~exist('fit_resEi400','var')
    %fit_data_file = 'FitEn_cut200_1Dsbg_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
    %fit_data_file =
'FitEn_cut200_1D_fit_bg_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-exponent fit
fixed bg at high momentum.

    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi400')
        fit_resEi400 = ld.fit_resEi400;
    else
        fit_resEi400 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei400meV.mat';
if ~exist('cuts2fit400','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit400 = ld.cuts2fit400;
end

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
cl = cuts2fit400.cutsF_list;
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end
cuts2fit400.cut110_sel = cl{1};
cuts2fit400.cut200_sel = cl{2}

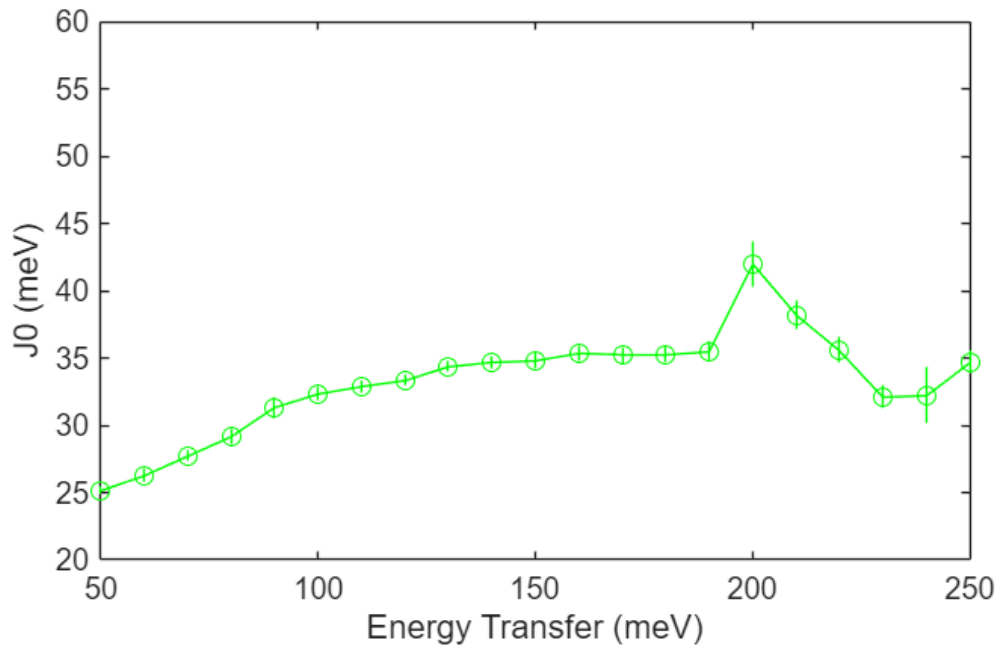
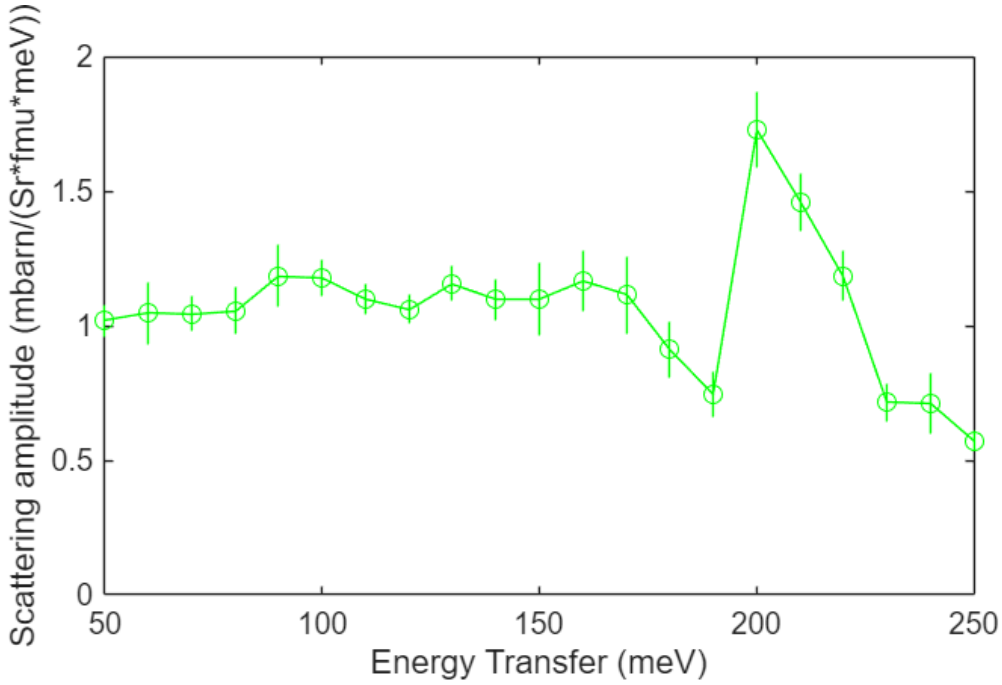
cuts2fit400 = struct with fields:
    cuts_list: {[1x1 sqw] [1x1 sqw] [1x1 sqw]}
    mask_ds2: {[2x13 double] [2x9 double]}
    mask_ds3: {[2x9 double] [2x5 double]}
    mask_ds1: {[2x13 double] [2x7 double]}
    bg_par_010off100: [1x1 struct]
    bg_par_010off200: [1x1 struct]
    bg_par_010off000: [1x1 struct]
    msk_of100_dir010: {[8x2 double] [18x2 double]}
    cutsF_list: {[1x1 sqw] [1x1 sqw] [1x1 sqw]}
    cut110_sel: [1x1 sqw]
    cutsS_list: {[1x1 sqw] [1x1 sqw] [1x1 sqw]}
    msk_of200_dir010: {[12x2 double] [9x2 double] [8x2 double]}

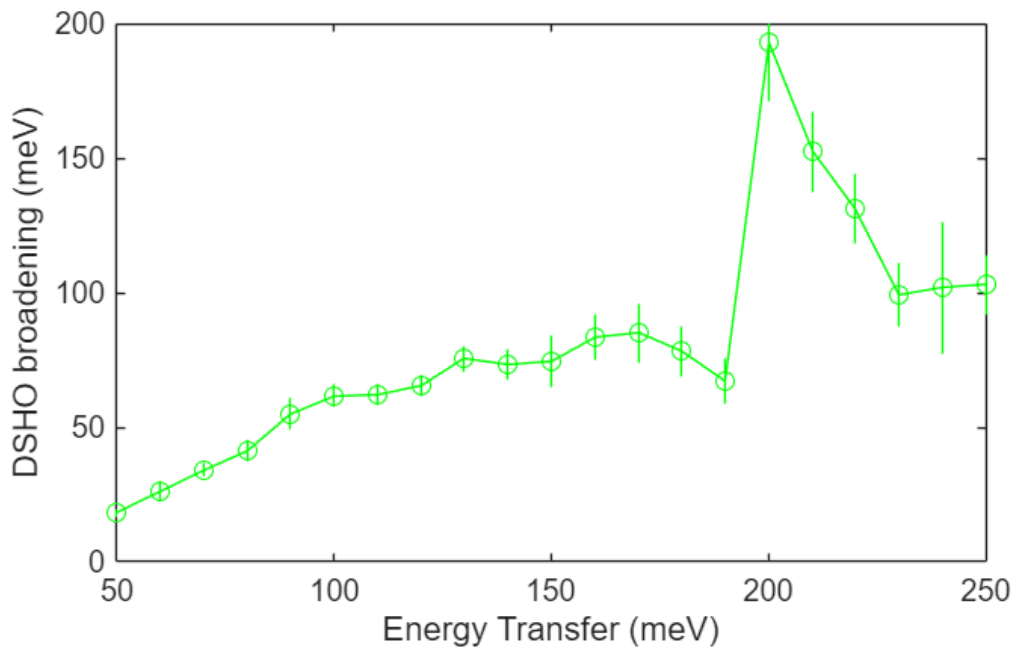
```

```
msk_of020_dir100: {[12x2 double] [10x2 double] [6x2 double]}
cut200_sel: [1x1 sqw]
```

```
cuts2fit = cuts2fit400;
fit_res = fit_resEi400;
```

```
[Sgn1,J0,G] = plot_SJG_results(fit_res,'g',{ 'S', 'J0', 'gamma'},{[0,2],[20,60],
[0,200]});
```





```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1))
```

```
dE =  
10
```

```
e_min = Sgn1.x(1);
```

```
En = 250
```

```
En =  
250
```

```
fit_par = fit_res.all_fit_par(1);  
dE_step = fit_par.en_range(2); %original energy transfer step data were binned  
to. No point in going finer  
half_dE = 0.5*(fit_par.en_range(3)-fit_par.en_range(1));  
en_idx = floor((En-e_min)/dE)+1;  
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);  
%the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[ ])};%(1:2);  
the2Dcuts = {cuts2fit.cut200_sel};  
  
%init_bg_params =  
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg  
%in case of 2-D fitting  
%  
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p  
arams,init_bg_params,false)  
lfit_par = fit_res.all_fit_par(en_idx);  
en_range = lfit_par.en_range;  
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.3  
5,true,false);
```

Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.8152

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 269 pixels -- processing data... -----> included 0 pixels

Number of points in input object: 450804
 Fraction of object selected: 0.06% (= 269 points) ; Access speed: 0.05GB/sec
 Fraction of object retained: 0.00% (= 0 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 5.7%; Transf. time frac.: 85.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 450804 pixels -- processing data... -----> included 372205 pixels

Number of points in input object: 450804
 Fraction of object selected: 100.00% (= 450804 points) ; Access speed: 1.26GB/sec
 Fraction of object retained: 82.56% (= 372205 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 60.0%; Transf. time frac.: 33.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 220530 pixels -- processing data... -----> retained 209459 pixels

Number of points in input object: 456366
 Fraction of object selected: 48.32% (= 220530 points) ; Access speed: 1.08GB/sec
 Fraction of object retained: 45.90% (= 209459 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 24.0%; Transf. time frac.: 34.8%; Resulting pix preparation: 41.1%

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Beginning fit (max 20 iterations)-----

Starting point

Total time = 2.3534s Reduced Chi^2 = 1.3828
 Free parameter values:
 103 0.5719 34.69 0.01835 -0.03409

Iteration = 1

Total time = 11.1594s Reduced Chi^2 = 1.5335 Levenberg-Marquardt = 0.1
 Free parameter values:
 83.62 0.4496 34.02 24.54 -3.201

Total time = 13.4022s Reduced Chi^2 = 1.3888 Levenberg-Marquardt = 1
 Free parameter values:
 96.29 0.532 34.51 7.887 -1.024

Total time = 15.6151s Reduced Chi^2 = 1.3815 Levenberg-Marquardt = 10
 Free parameter values:
 100.8 0.5594 34.64 2.524 -0.3437

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	100.8	+/-	NaN
4	0.5594	+/-	NaN
5	0		
6	34.64	+/-	NaN
7	0		

8	0
9	0
10	0

Background parameter values:

```
-----
1      2.524   +/-   NaN
2     -0.3437  +/-   NaN
3     -0.006085
4     -0.0001965
```

Covariance matrix for free parameters:

```
-----
NaN  NaN  NaN  NaN  NaN
NaN  NaN  NaN  NaN  NaN
NaN  NaN  NaN  NaN  NaN
NaN  NaN  NaN  NaN  NaN
NaN  NaN  NaN  NaN  NaN
```

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 456366 pixels -- processing data... -----> included 380948 pixels

```
-----
Number of points in input object: 456366
Fraction of object selected: 100.00% (= 456366 points) ; Access speed: 2.17GB/sec
Fraction of object retained: 83.47% (= 380948 points)
```

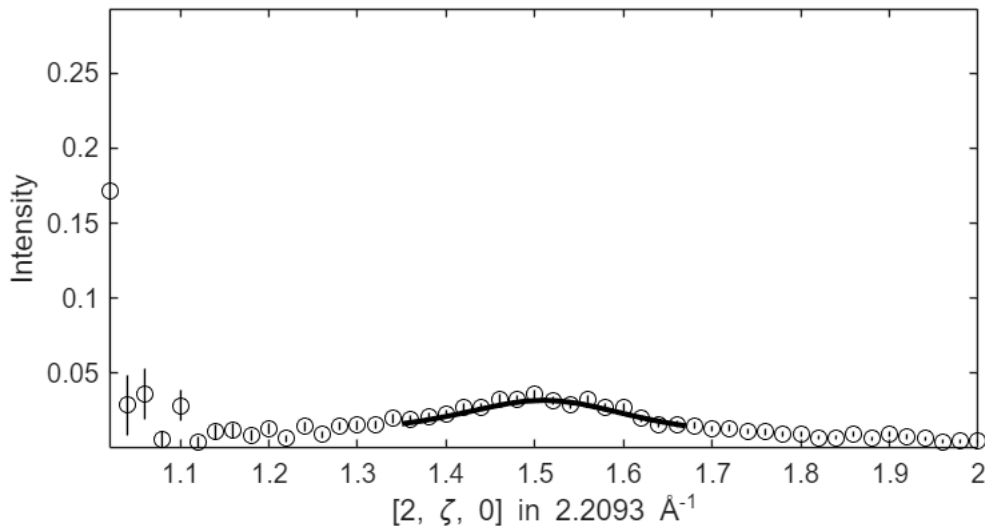
**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 51.0%; Transf. time frac.: 47.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 209401 pixels -- processing data... -----> included 185706 pixels

```
-----
Number of points in input object: 209401
Fraction of object selected: 100.00% (= 209401 points) ; Access speed: 2.00GB/sec
Fraction of object retained: 88.68% (= 185706 points)
```

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 44.8%; Transf. time frac.: 51.8%; Resulting pix preparation: 0.0%

Fe_ei401_align_sym3.sqw
 S=0.56± NaN; J0= 35± NaN; gamma=100.774±NaN
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $240 \leq E \leq 260$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 456366 pixels -- processing data... -----> retained 380948 pixels

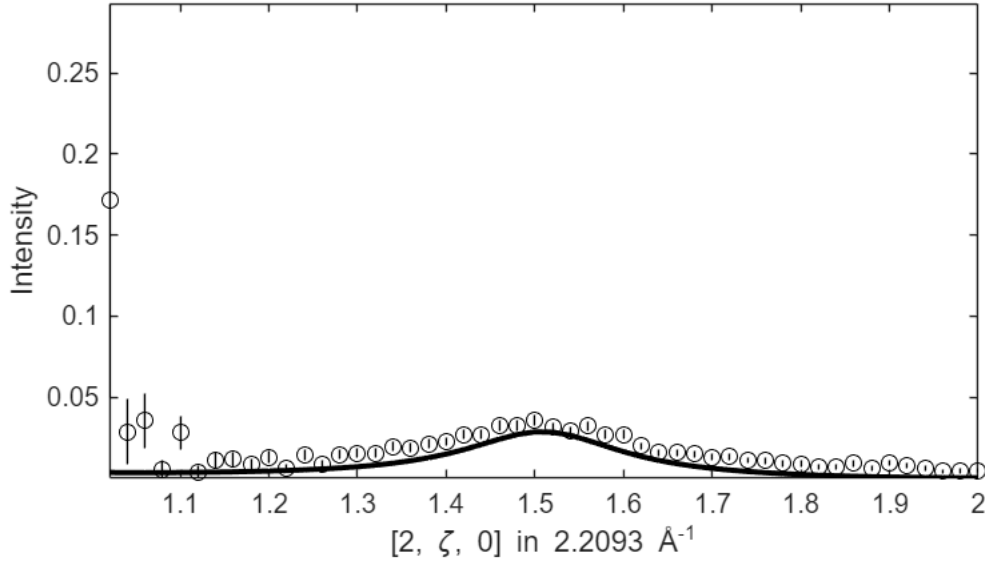
```
-----
Number of points in input object: 456366
```

Fraction of object selected: 100.00% (= 456366 points) ; Access speed: 2.11GB/sec
 Fraction of object retained: 83.47% (= 380948 points)

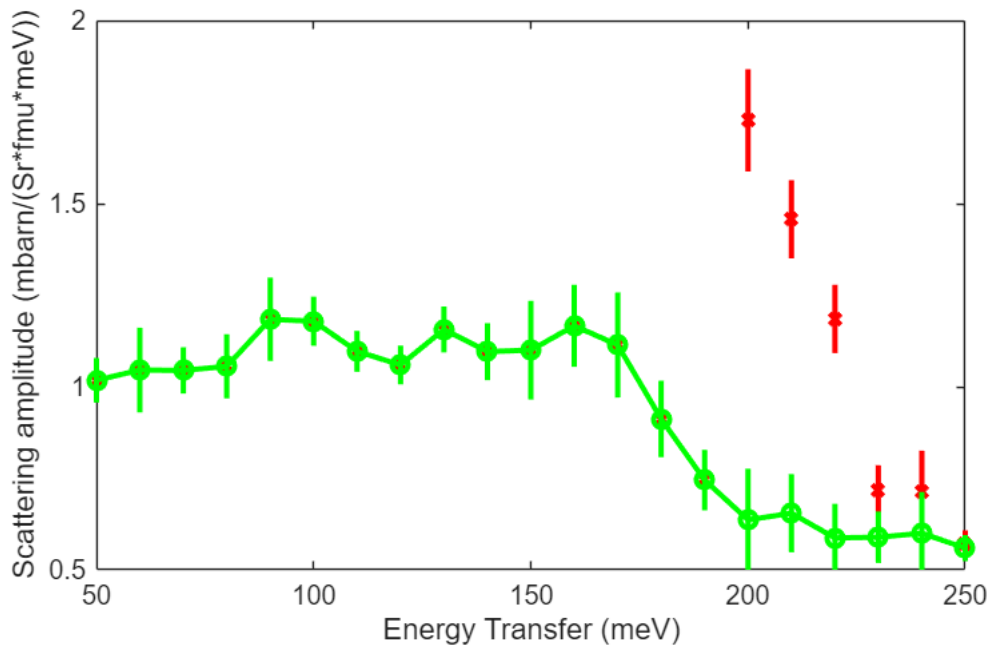
**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 25.9%; Transf. time frac.: 57.2%; Resulting pix preparation: 16.9%

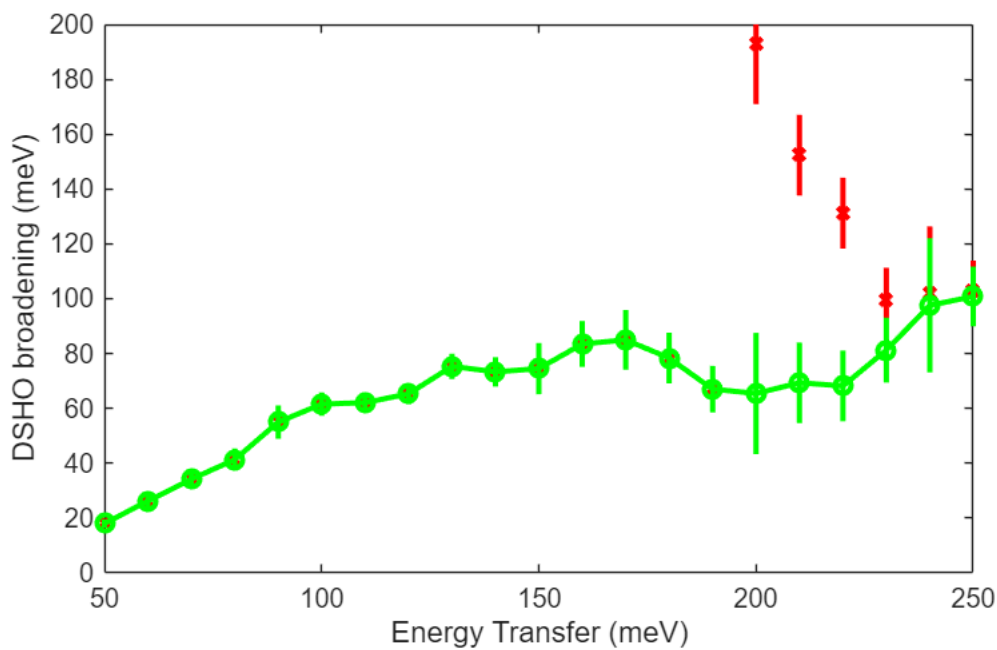
Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $240 \leq E \leq 260$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



```
%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
Sgn1.signal(en_idx) = fit_par.p(4);
J0.signal(en_idx) = fit_par.p(6);
G.signal(en_idx) = fit_par.p(3);
plot_SJG_results(fit_res, 'g', {'S', 'J0', 'gamma'}, {[0.5, 2], [20, 60], [0, 200]},
{Sgn1, J0, G});
```





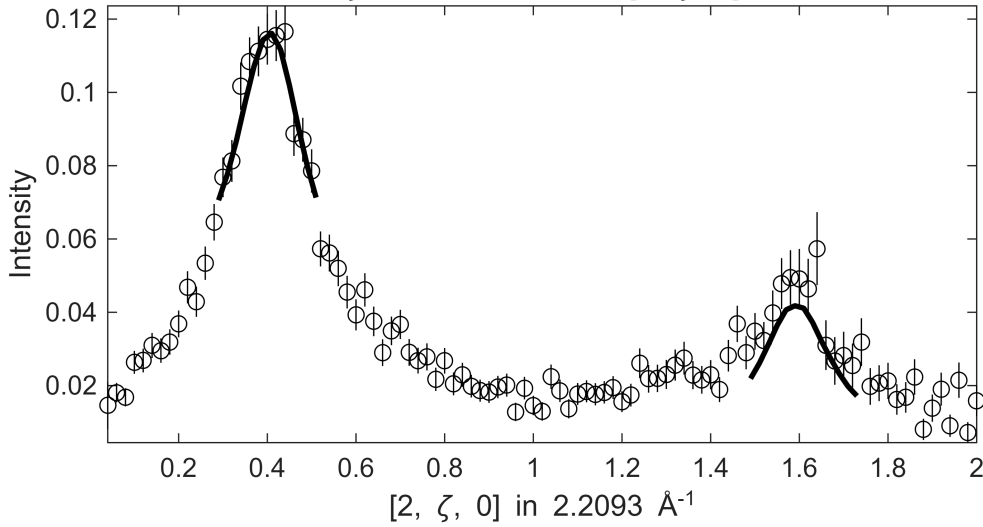
$$\begin{array}{c} -0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 160 \leq E \leq 180 \\ \zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0] \end{array}$$

$$\begin{array}{c} -0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 170 \leq E \leq 190 \\ \zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0] \end{array}$$

Fe_ei401_align_sym3.sqw

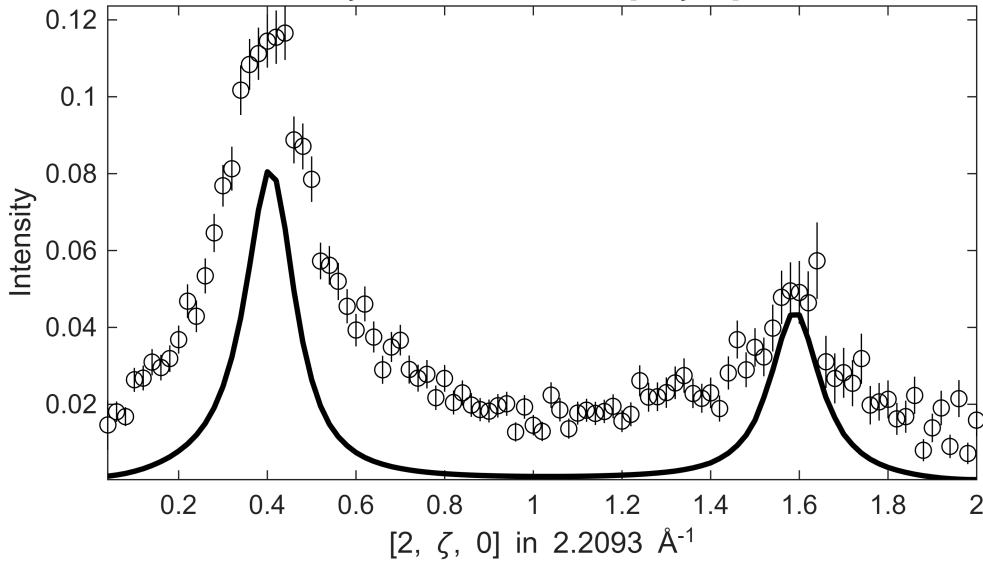
S=0.51±0.26; J0= 34± 1.1; gamma=55.5442±23.3527

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 180 ≤ E ≤ 200
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 180 ≤ E ≤ 200
ζ=-0.01:0.02:2.01 in [2, ζ, 0]



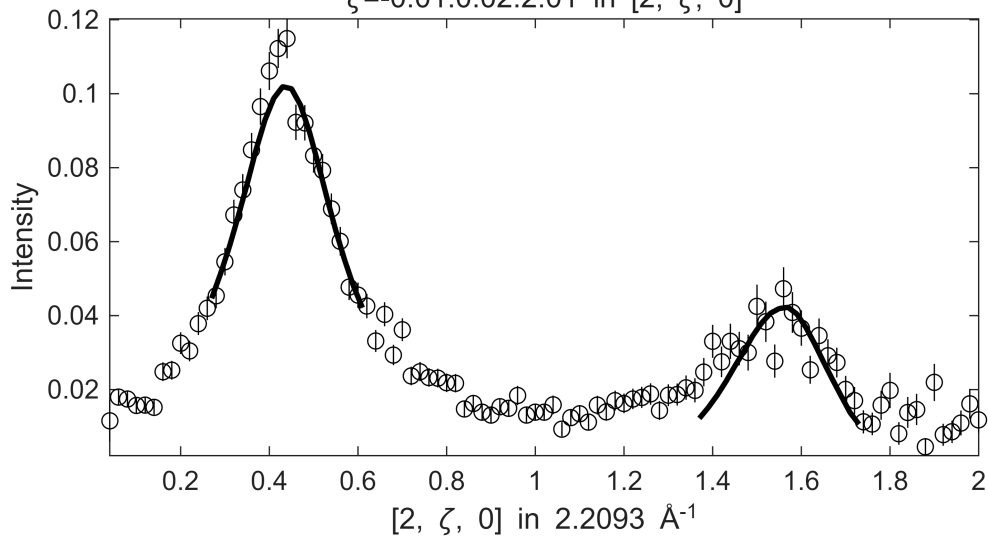
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 4#9 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:12 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802
Warning: Dataset 2:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.83333

$$\begin{array}{c} -0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 190 \leq E \leq 210 \\ \zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0] \end{array}$$

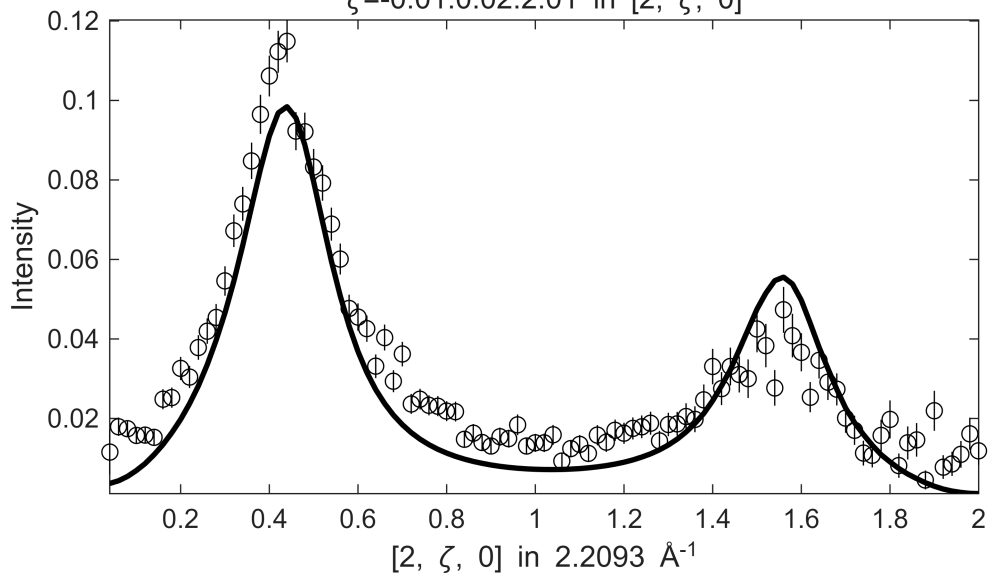
S= 1.4±0.43; J0= 37± 2.1; gamma=135.098±34.2172

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 200 ≤ E ≤ 220
 0.12 ζ=-0.01:0.02:2.01 in [2, ζ, 0]



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $200 \leq E \leq 220$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$

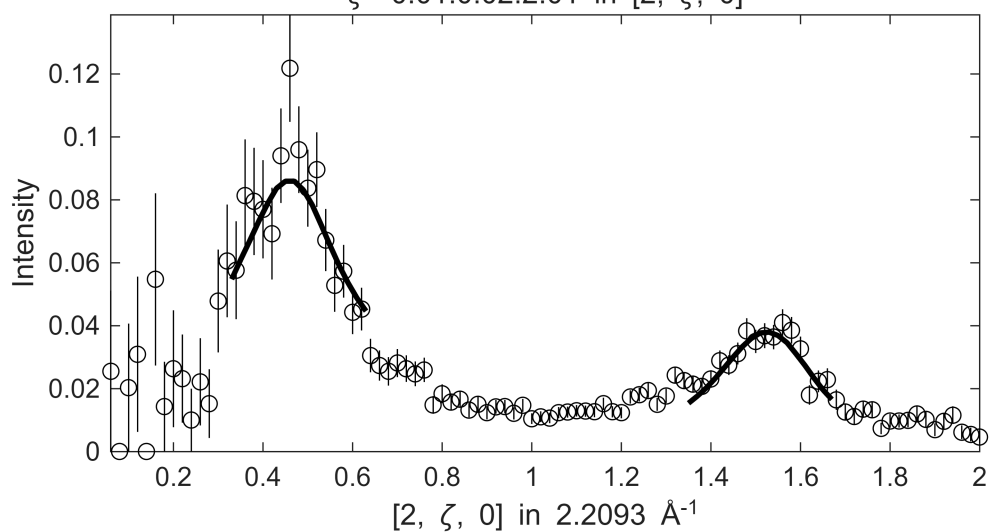
[illegible]

```
Warning: Dataset:18 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 1:2 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.8519
```

$$\begin{array}{c} -0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 210 \leq E \leq 230 \\ \zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0] \end{array}$$

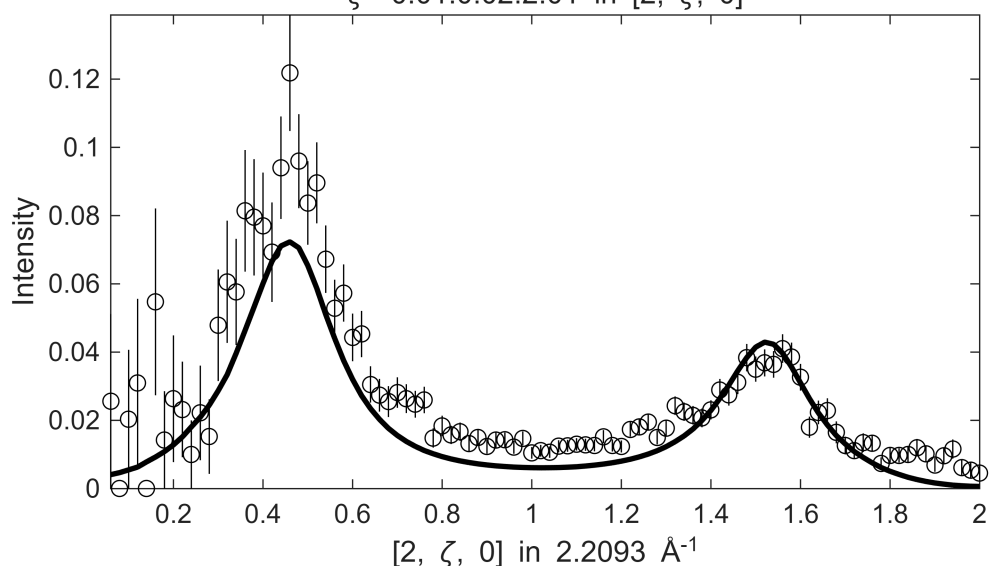
$S=0.99\pm0.68$; $J_0=35\pm3.7$; $\gamma=121.232\pm65.0773$

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $220 \leq E \leq 240$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



Fe_ei401_align_sym3.sqw

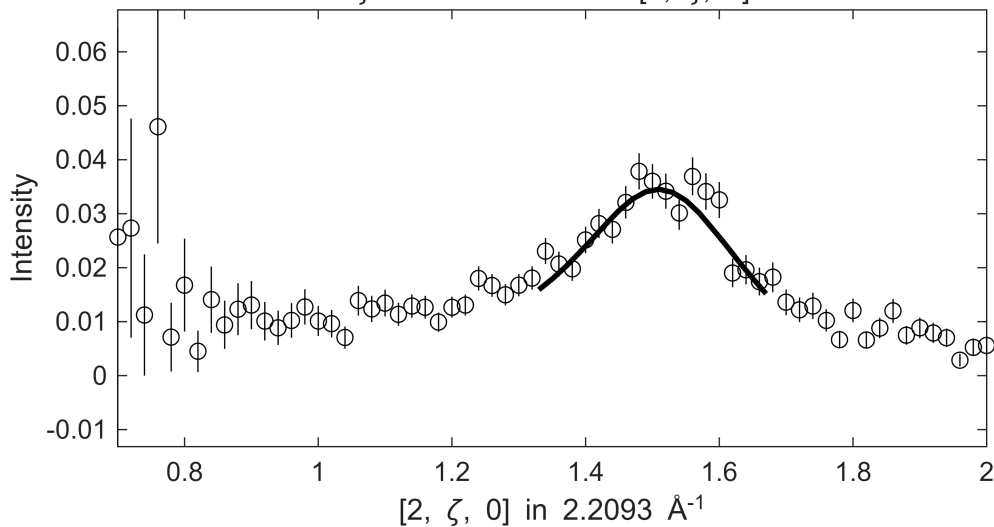
$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $220 \leq E \leq 240$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$

[illegible]

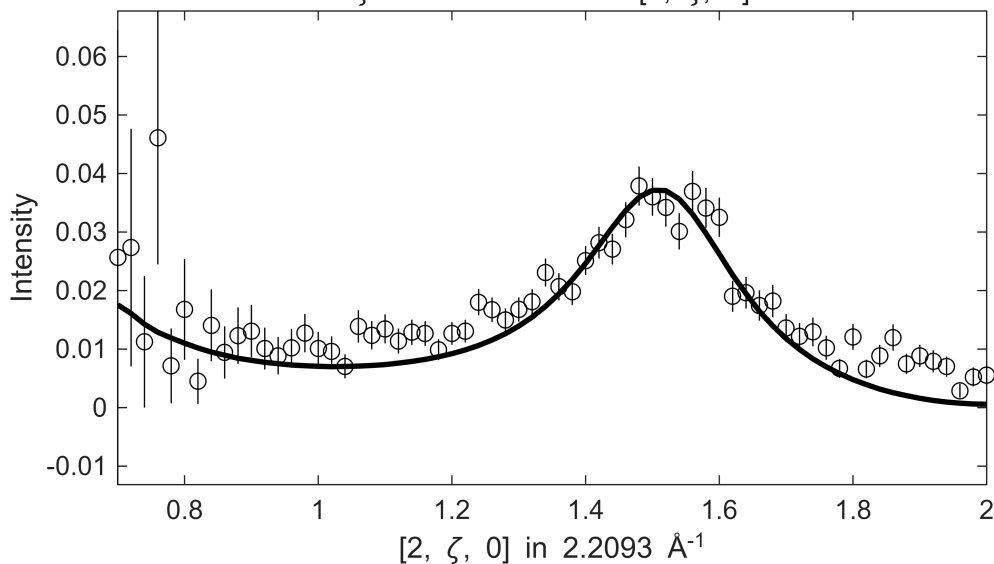
Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.8152

Fe_ei401_align_sym3.sqw

S= 1 $\pm$ 0.026; J0= 36 $\pm$ 0.81; gamma=148.148 $\pm$ 11.0337

$$-0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 230 \leq E \leq 250$$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$ 

Fe_ei401_align_sym3.sqw

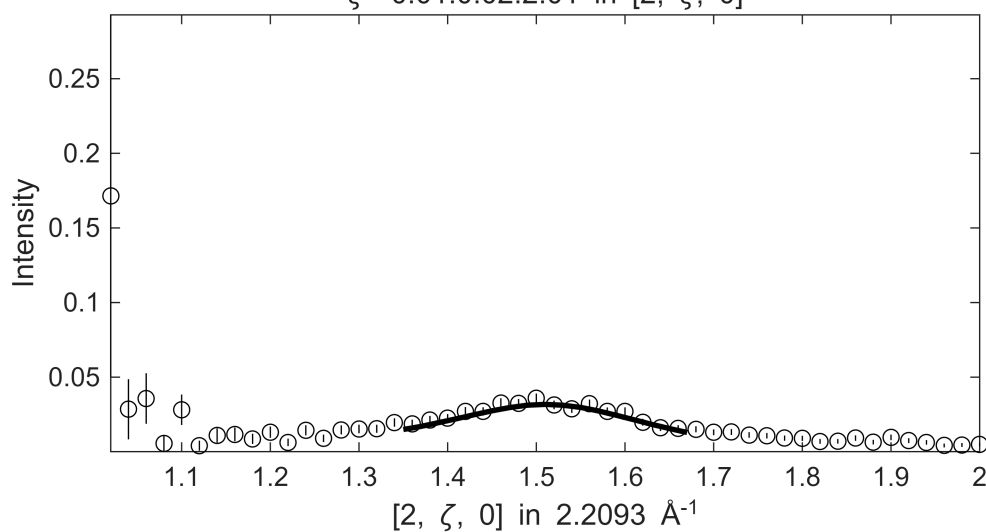
$$-0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 230 \leq E \leq 250$$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$ [illegible]

Warning: Dataset:11 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.8152

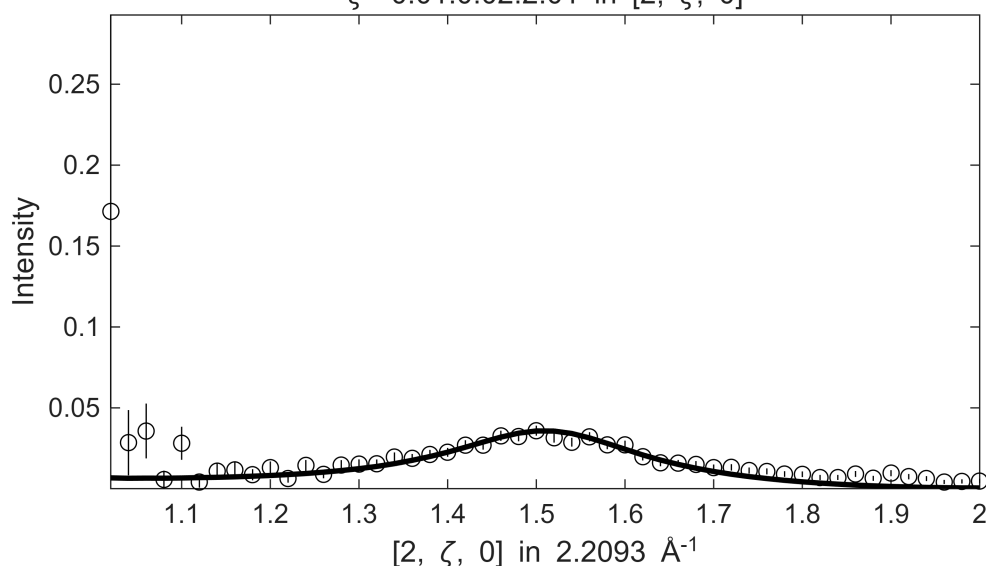
```
Warning: Dataset: 2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.9608
```

Fe_ei401_align_sym3.sqw

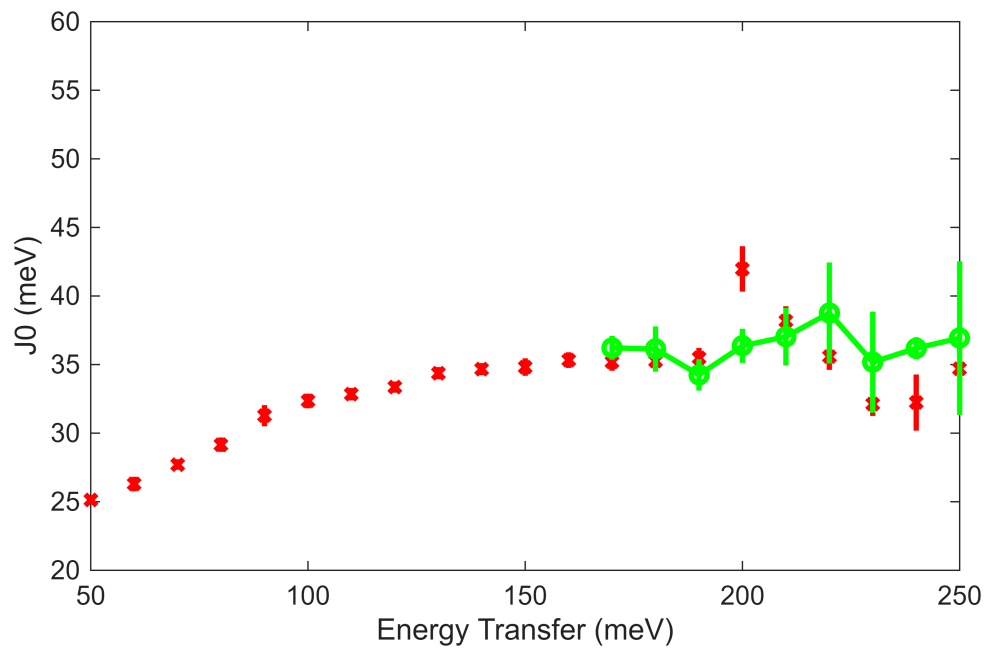
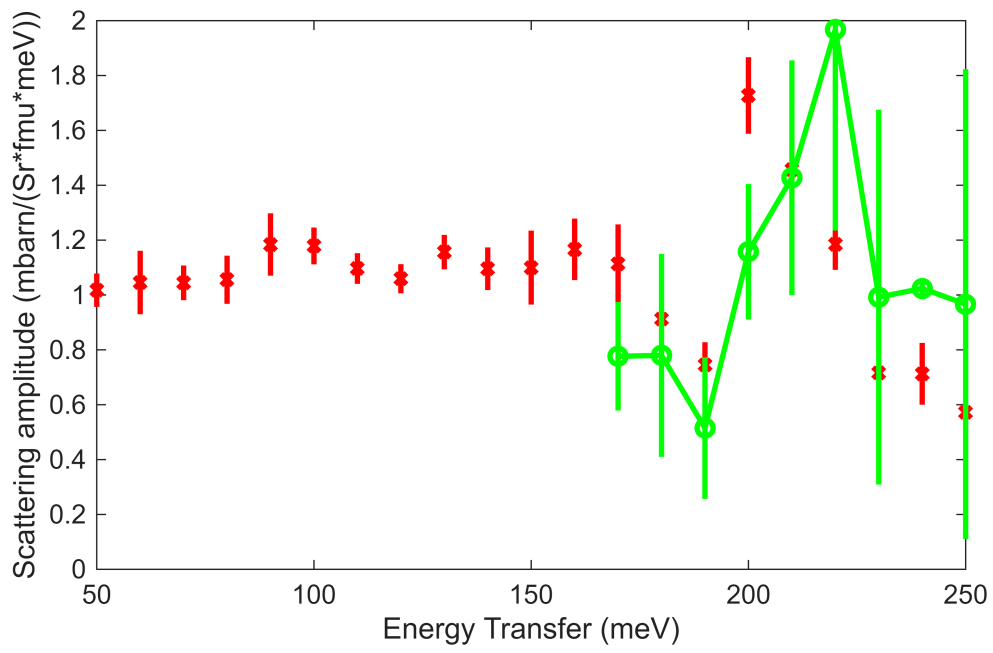
$S=0.97\pm0.86$; $J_0=37\pm5.6$; $\gamma=143.963\pm99.3441$

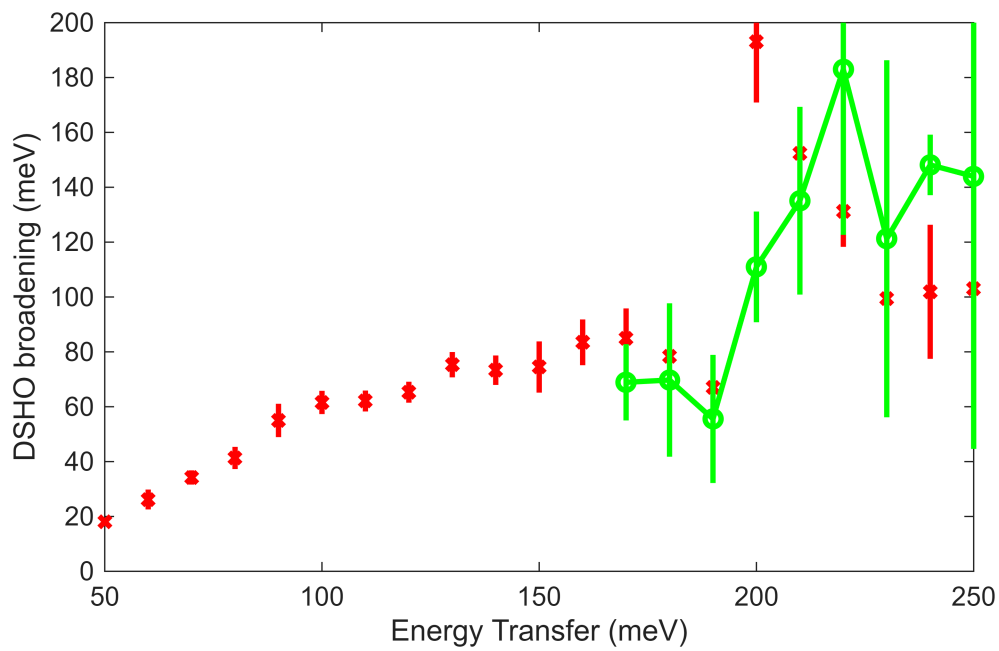
$$-0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 240 \leq E \leq 260$$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$ 

Fe_ei401_align_sym3.sqw

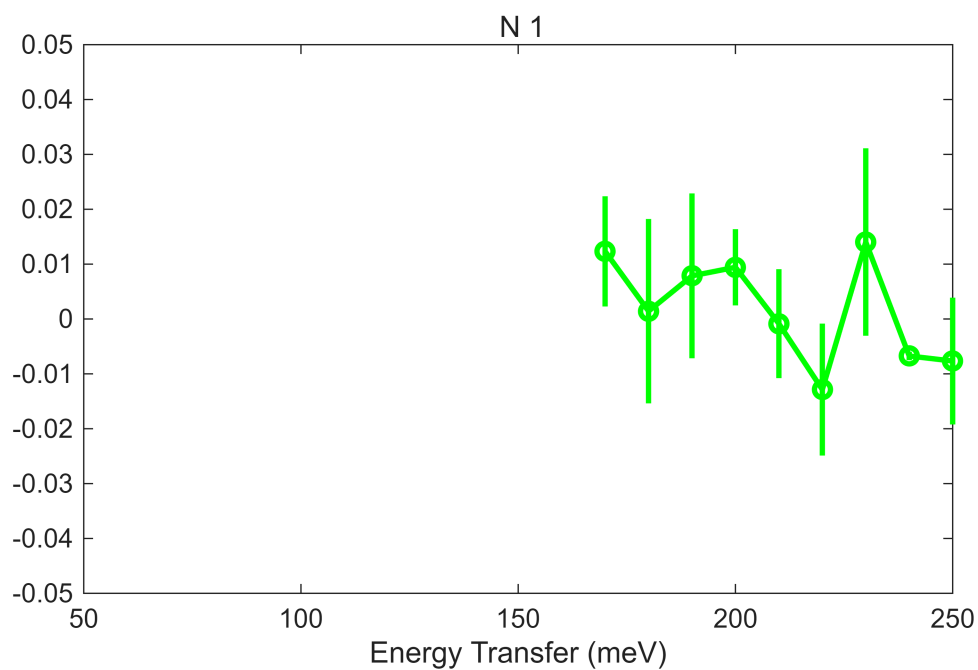
$$-0.1 \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], \quad -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], \quad 240 \leq E \leq 260$$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$ [illegible]

```
%plot_SJG_results(fit_res, 'g', {'S', 'J0', 'gamma'}, {[0,2],[20,60],[0,200]},  
{Sg,J0,G});  
plot_SJG_results(fit_res2peak, 'g', {'S', 'J0', 'gamma'}, {[0,2],[20,60],[0,200]},  
{Sg,J0,G});
```

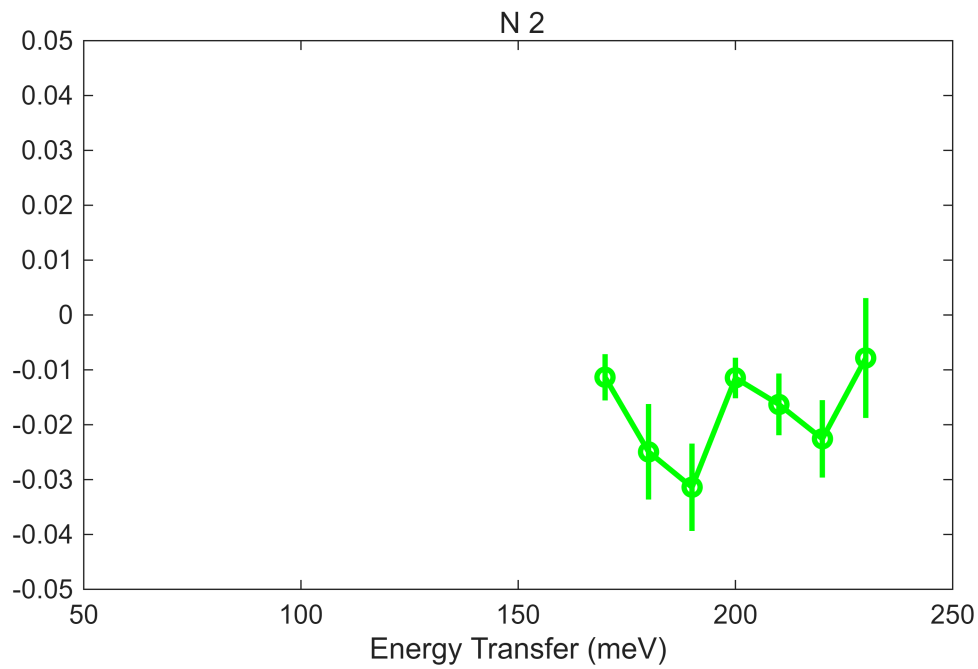





```
bg_data = extract_bg_par(fit_res2peak.peak_fit_par,5);
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure;ly -0.05 0.05
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
igure;ly -0.05 0.05;
```



```
%pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;ly -5 5;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;ly -10 5;
```

```
fit_res.Speak_varsl = Sg;
fit_res.J0peak_varsl = J0;
fit_res.gamma_peak_varsl = G;
% fit_res.Speak = Sg;
% fit_res.J0peak = J0;
% fit_res.gamma_peak = G;
%fit_res.peak_fit_1_2peaksFixSlope_par =
[fit_res1peak.peak_fit_par(:),fit_res1peak.peak_fit_par(:)];
fit_resEi400 = fit_res;
%save(fullfile(fit_data_path,fit_data_file),"fit_resEi400");
```